



Qualification Pack



Assistant Machinist - Iron and Steel

QP Code: ISC/Q0909

Version: 5.0

NSQF Level: 3

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ISC/Q0909: Assistant Machinist - Iron and Steel

Brief Job Description

This individual is primarily involved in various machining and inspection work on conventional/manual machines such as quality verification, minor repair work, change of worn out tools, re-setting of the tools, etc.

Personal Attributes

The person should be patient, organised, team-oriented and have the ability to work for long hours in adverse conditions. They must be able to plan and prioritise tasks effectively and have an eye for detail and quality.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ISC/N0008: Use basic health and safety practices at the workplace](#)
2. [ISC/N0009: Work effectively with others](#)
3. [ISC/N0901: Prepare for fabrication activities](#)
4. [ISC/N0902: Perform fabrication activities](#)
5. [ISC/N0903: Perform post-fabrication activities](#)
6. [ISC/N0904: Perform maintenance activities](#)
7. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Iron and Steel
Sub-Sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory, Iron Making
Occupation	Mechanical Maintenance
Country	India
NSQF Level	3



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Credits	14
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7233.0102
Minimum Educational Qualification & Experience	10th Class OR 9th Class with 1.5 years of experience in relevant field OR Grade 8 pass with 2 years of (NTC/ NAC) after 8th OR 8th Class with 3 Years of experience in relevant field OR 5th Class
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	01/04/2028
NSQC Approval Date	01/04/2025
Version	5.0
Reference code on NQR	QG-03-IS-03879-2025-V2-ISSSC
NQR Version	1



Qualification Pack

ISC/N0008: Use basic health and safety practices at the workplace

Description

This OS unit is about following safety and adopting sustainable practices for optimising use of resources.

Scope

The scope covers the following :

- Maintain safe and secure working environment
- Emergencies, rescue and first aid procedures
- Health and hygiene
- Housekeeping and waste management
- Material and energy conservation

Elements and Performance Criteria

Maintain safe and secure working environment

To be competent, the user/individual on the job must be able to:

- PC1.** identify hazardous activities and the possible causes of risks or accidents in the workplace
- PC2.** follow safe working practices while dealing with hazards to ensure safety of self and others
- PC3.** use appropriate protective clothing/equipment for specific tasks and work
- PC4.** follow appropriate safety practices while working in and around trenches, elevated places and confined areas
- PC5.** lift heavy objects safely using correct procedures
- PC6.** carry out routine check of the machine for identifying potential hazards
- PC7.** report any identified breaches in health, safety and security policies and procedures to the designated person

Emergencies, rescue and first aid procedures

To be competent, the user/individual on the job must be able to:

- PC8.** use appropriate type of fire extinguisher
- PC9.** apply appropriate rescue techniques during fire hazard
- PC10.** provide appropriate first aid procedure to victims wherever required eg.in case of bleeding, burns, choking, electric shock etc.
- PC11.** follow emergency procedures such as raising alarm, safe evacuation etc.
- PC12.** attend safety training and fire drills to respond promptly during an emergency

Health and hygiene

To be competent, the user/individual on the job must be able to:

- PC13.** follow regular cleaning and disinfection practices at work place using appropriate techniques and materials
- PC14.** follow hand hygiene practices at work place using appropriate techniques and materials
- PC15.** report regarding the contagious illness of self or people in close contact
- PC16.** avoid contact with ill people and self-isolate in a similar situation



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Housekeeping and waste management

To be competent, the user/individual on the job must be able to:

- PC17.** follow the fundamentals of 5S for housekeeping
- PC18.** ensure good housekeeping in order to prevent hazards and accidents
- PC19.** store the material, tools and equipment in the correct location and in good condition
- PC20.** segregate waste into different categories
- PC21.** identify recyclable, non-recyclable and hazardous waste
- PC22.** dispose non-recyclable, recyclable and reusable waste appropriately at identified location

Material and energy conservation

To be competent, the user/individual on the job must be able to:

- PC23.** identify ways to optimize usage of material in various tasks/activities/processes
- PC24.** check for spills/leakages in various tasks/activities/processes
- PC25.** plug spills/leakages and escalate to appropriate authority if unable to rectify
- PC26.** check if the equipment/machine is functioning normally before commencing work and rectify wherever required
- PC27.** ensure electrical equipment and appliances are properly connected and turned off when not in use

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organisation procedures for health, safety and security, individual role and responsibilities in this context
- KU2.** the organisation's emergency procedures for different emergency situations and the importance of following the same
- KU3.** evacuation procedures for workers and visitors
- KU4.** how and when to report hazards
- KU5.** potential hazards, risks and threats based on the nature of work
- KU6.** preventative and remedial actions to be taken in case of exposure to toxic material
- KU7.** various types of fire extinguisher
- KU8.** various types of safety signs and their meaning
- KU9.** appropriate first aid treatment relevant to different condition e.g. bleeding, minor burns, eye injuries etc.
- KU10.** relevant standards, procedures and policies related to 5S followed in the company
- KU11.** the various materials used and their storage norms
- KU12.** efficient utilisation of material and water
- KU13.** basics of electricity and prevalent energy efficient devices
- KU14.** common practices of conserving electricity
- KU15.** categorisation of waste into dry, wet, recyclable, non-recyclable and items of single-use plastics
- KU16.** usage of different colors of dustbins



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KU17. waste management techniques

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read safety instructions/guidelines
- GS2.** modify work practices to improve them
- GS3.** ask for clarifications from superior about the job requirement
- GS4.** work with supervisors/team members to carry out work related tasks
- GS5.** complete tasks efficiently and accurately within stipulated time
- GS6.** inform/report to concerned person in case of any problem
- GS7.** make timely decisions for efficient utilization of resources
- GS8.** write reports such as accident report, in at least English/regional language
- GS9.** be punctual and utilize time efficiently



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain safe and secure working environment</i>	10	14	-	6
PC1. identify hazardous activities and the possible causes of risks or accidents in the workplace	2	2	-	1
PC2. follow safe working practices while dealing with hazards to ensure safety of self and others	2	3	-	1
PC3. use appropriate protective clothing/equipment for specific tasks and work	1	2	-	1
PC4. follow appropriate safety practices while working in and around trenches, elevated places and confined areas	2	1	-	-
PC5. lift heavy objects safely using correct procedures	1	2	-	1
PC6. carry out routine check of the machine for identifying potential hazards	1	2	-	1
PC7. report any identified breaches in health, safety and security policies and procedures to the designated person	1	2	-	1
<i>Emergencies, rescue and first aid procedures</i>	6	9	-	5
PC8. use appropriate type of fire extinguisher	1	1	-	1
PC9. apply appropriate rescue techniques during fire hazard	1	2	-	1
PC10. provide appropriate first aid procedure to victims wherever required eg.in case of bleeding, burns, choking, electric shock etc.	2	2	-	1
PC11. follow emergency procedures such as raising alarm, safe evacuation etc.	1	2	-	1
PC12. attend safety training and fire drills to respond promptly during an emergency	1	2	-	1
<i>Health and hygiene</i>	2	6	-	2



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. follow regular cleaning and disinfection practices at work place using appropriate techniques and materials	1	2	-	1
PC14. follow hand hygiene practices at work place using appropriate techniques and materials	1	2	-	1
PC15. report regarding the contagious illness of self or people in close contact	-	1	-	-
PC16. avoid contact with ill people and self-isolate in a similar situation	-	1	-	-
<i>Housekeeping and waste management</i>	7	12	-	5
PC17. follow the fundamentals of 5S for housekeeping	2	3	-	2
PC18. ensure good housekeeping in order to prevent hazards and accidents	1	2	-	-
PC19. store the material, tools and equipment in the correct location and in good condition	1	2	-	-
PC20. segregate waste into different categories	1	2	-	1
PC21. identify recyclable, non-recyclable and hazardous waste	1	1	-	1
PC22. dispose non-recyclable, recyclable and reusable waste appropriately at identified location	1	2	-	1
<i>Material and energy conservation</i>	5	9	-	2
PC23. identify ways to optimize usage of material in various tasks/activities/processes	1	2	-	-
PC24. check for spills/leakages in various tasks/activities/processes	1	2	-	1
PC25. plug spills/leakages and escalate to appropriate authority if unable to rectify	1	2	-	1
PC26. check if the equipment/machine is functioning normally before commencing work and rectify wherever required	1	2	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. ensure electrical equipment and appliances are properly connected and turned off when not in use	1	1	-	-
NOS Total	30	50	-	20



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National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0008
NOS Name	Use basic health and safety practices at the workplace
Sector	Iron and Steel
Sub-Sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory
Occupation	Mechanical Maintenance, Raw Material Handling , Product Quality Control, Refractory Maintenance, Furnace Operation, Fitter, Cold Rolling Mill Operations, Electrical Maintenance, Mechanical Maintenance
NSQF Level	3
Credits	TBD
Version	2.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ISC/N0009: Work effectively with others

Description

This OS unit is about communicating with colleagues/superiors and others, either in own work group or in other work groups within organisation.

Scope

The scope covers the following :

- Communicate effectively with colleagues and others
- Interact with supervisor
- Follow appropriate behaviour at work place

Elements and Performance Criteria

Communicate effectively with colleagues and others

To be competent, the user/individual on the job must be able to:

- PC1.** coordinate with colleagues to share work, as per the workload in order to achieve team goals
- PC2.** maintain clear communication with colleagues and others, wherever needed, through all means i.e. face-to-face, telephonic or written
- PC3.** adjust communication styles to reflect gender and persons with disability (PwD) sensitivity
- PC4.** respect all colleagues and co-workers
- PC5.** resolve conflicts by communicating with colleagues and other departments

Interact with supervisor

To be competent, the user/individual on the job must be able to:

- PC6.** identify work requirements by receiving instructions from reporting supervisor
- PC7.** escalate problems to supervisors that cannot be handled
- PC8.** report the completed work
- PC9.** interact with the reporting supervisor about any possible hazards and safety concerns

Follow appropriate behaviour at work place

To be competent, the user/individual on the job must be able to:

- PC10.** extend help to people with Disability (PwD) at workplace, if required
- PC11.** empathize with people with disability
- PC12.** adopt a gender neutral behavior
- PC13.** adopt responsible and disciplined behaviours at the workplace

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the importance of effective communication and establishing good working relationships with colleagues and supervisor



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- KU2.** different methods of communication as per the circumstances
- KU3.** importance of teamwork in organization and individual success
- KU4.** various components of effective communication
- KU5.** barriers to effective communication
- KU6.** common reasons for interpersonal conflict
- KU7.** what constitutes disciplined behaviour for a working professional
- KU8.** gender concepts, issues & legislation
- KU9.** organisational policies and procedures related to gender equality
- KU10.** challenges faced by PWD and the ways to help them overcome the same

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read instructions/guidelines/procedures
- GS2.** listen effectively and orally communicate information
- GS3.** ask for clarification and advice from the concerned person
- GS4.** maintain positive and effective relationships with colleagues
- GS5.** evaluate the possible solution(s) to the problem
- GS6.** spot and communicate potential areas of disruptions in the work process and report the same
- GS7.** complete written work with attention to detail
- GS8.** check that the work meets customer requirements



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Communicate effectively with colleagues and others</i>	13	20	-	9
PC1. coordinate with colleagues to share work, as per the workload in order to achieve team goals	3	5	-	2
PC2. maintain clear communication with colleagues and others, wherever needed, through all means i.e. face-to-face, telephonic or written	5	7	-	3
PC3. adjust communication styles to reflect gender and persons with disability (PwD) sensitivity	3	4	-	2
PC4. respect all colleagues and co-workers	1	2	-	1
PC5. resolve conflicts by communicating with colleagues and other departments	1	2	-	1
<i>Interact with supervisor</i>	8	14	-	6
PC6. identify work requirements by receiving instructions from reporting supervisor	2	3	-	1
PC7. escalate problems to supervisors that cannot be handled	2	3	-	2
PC8. report the completed work	2	3	-	1
PC9. interact with the reporting supervisor about any possible hazards and safety concerns	2	5	-	2
<i>Follow appropriate behaviour at work place</i>	9	16	-	5
PC10. extend help to people with Disability (PwD) at workplace, if required	2	4	-	2
PC11. empathize with people with disability	2	4	-	1
PC12. adopt a gender neutral behavior	2	4	-	1
PC13. adopt responsible and disciplined behaviours at the workplace	3	4	-	1



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	50	-	20



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National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0009
NOS Name	Work effectively with others
Sector	Iron and Steel
Sub-Sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory
Occupation	Mechanical Maintenance, Raw Material Handling , Product Quality Control, Refractory Maintenance, Furnace Operation, Fitter, Operations, Coke Making, Mechanical Maintenance
NSQF Level	3
Credits	TBD
Version	2.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ISC/N0901: Prepare for fabrication activities

Description

This unit is about preparing for fabrication activities as per the given work instructions and the standards specified by the organisation.

Scope

The scope covers the following :

- Identify work requirements
- Prepare for fabrication work

Elements and Performance Criteria

Identify work requirements

To be competent, the user/individual on the job must be able to:

- PC1.** identify the work to be done by interpreting the engineering drawings/blueprints/SOPs
- PC2.** compute dimensions, sizes, shapes and tolerances of sub-assemblies of the machine based on the specifications as mentioned in the drawings/blueprints
- PC3.** report and rectify cases of any inappropriate information in design documents as per organizational procedures
- PC4.** identify the tools, dies, measuring instruments, machines (lathes, milling machines, grinders etc.) and input material (metal) required for the job
- PC5.** select and arrange the right input material, dies, tools, measuring instruments and machines as per the SOP and job requirements

Prepare for fabrication work

To be competent, the user/individual on the job must be able to:

- PC6.** use appropriate Personal Protective Equipment (PPE) for safe working in workshop
- PC7.** plan sequence of activities need to perform for fabrication work
- PC8.** check the tools, dies, measuring instruments and machines are cleaned, free from any defects and functioning properly
- PC9.** check and calibrate the tools and equipment before use
- PC10.** check that input material is cleaned i.e. free from rust or any contaminants and as per the required quality standard
- PC11.** ensure availability of correct quantity metal work pieces and other materials required for fabrication in the workshop
- PC12.** smoothen out the metal work piece prior to fabrication by using grinding machine
- PC13.** measure and mark reference points/cutting lines on the work pieces by using compass, callipers, rulers and other measuring tools
- PC14.** set the conventional or computer numerically controlled machines and its parameters as per the design requirements and SOP

Knowledge and Understanding (KU)



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The individual on the job needs to know and understand:

- KU1.** relevant standards and procedures followed in the company
- KU2.** importance of identifying non-conforming material and storage of the same
- KU3.** risk and impact of not following defined procedures/work instructions
- KU4.** various metal fabrication processes such as forging, rolling, extrusion, bending, spinning, drawing, drilling, sawing, boring, threading, grinding, cutting, etc.
- KU5.** SOP recommended by the manufacturer for using tools, jigs, fixtures, measuring instruments etc., required during the fabrication work
- KU6.** the impact of various machining parameters on the final product
- KU7.** the use of various cutting tools for different machining operations
- KU8.** how to read engineering drawings and their symbols
- KU9.** fundamentals of the CNC/conventional machine
- KU10.** SOP recommended by the organisation for operating CNC and conventional machine
- KU11.** the metal properties and metallurgy
- KU12.** safety requirements for CNC and conventional machine during the machining work

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read work instructions, equipment manuals and process documents
- GS2.** communicate the process requirements to the supervisor and co-workers
- GS3.** attentively listen and comprehend the information given by the supervisor/team members
- GS4.** write work related information in English/regional language
- GS5.** recognise a workplace problem and take suitable action
- GS6.** analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently
- GS7.** plan and organise work according to the work requirements
- GS8.** complete the assigned tasks with minimum supervision
- GS9.** report to the supervisor or deal with a colleague individually, depending on the type of concern



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Identify work requirements</i>	11	14	-	9
PC1. identify the work to be done by interpreting the engineering drawings/blueprints/SOPs	1	2	-	1
PC2. compute dimensions, sizes, shapes and tolerances of sub-assemblies of the machine based on the specifications as mentioned in the drawings/blueprints	2	3	-	2
PC3. report and rectify cases of any inappropriate information in design documents as per organizational procedures	1	2	-	1
PC4. identify the tools, dies, measuring instruments, machines (lathes, milling machines, grinders etc.) and input material (metal) required for the job	5	3	-	3
PC5. select and arrange the right input material, dies, tools, measuring instruments and machines as per the SOP and job requirements	2	4	-	2
<i>Prepare for fabrication work</i>	19	36	-	11
PC6. use appropriate Personal Protective Equipment (PPE) for safe working in workshop	1	2	-	1
PC7. plan sequence of activities need to perform for fabrication work	1	2	-	-
PC8. check the tools, dies, measuring instruments and machines are cleaned, free from any defects and functioning properly	3	6	-	2
PC9. check and calibrate the tools and equipment before use	2	5	-	1
PC10. check that input material is cleaned i.e. free from rust or any contaminants and as per the required quality standard	3	5	-	2
PC11. ensure availability of correct quantity metal work pieces and other materials required for fabrication in the workshop	1	2	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. smoothen out the metal work piece prior to fabrication by using grinding machine	2	4	-	1
PC13. measure and mark reference points/cutting lines on the work pieces by using compass, callipers, rulers and other measuring tools	2	4	-	1
PC14. set the conventional or computer numerically controlled machines and its parameters as per the design requirements and SOP	4	6	-	3
NOS Total	30	50	-	20



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National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0901
NOS Name	Prepare for fabrication activities
Sector	Iron and Steel
Sub-Sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory
Occupation	Mechanical Maintenance
NSQF Level	3
Credits	TBD
Version	2.0
Last Reviewed Date	01/04/2025
Next Review Date	01/04/2028
NSQC Clearance Date	01/04/2025



Qualification Pack

ISC/N0902: Perform fabrication activities

Description

This unit is about performing fabrication of components to ensure that the final products manufactured is as per the quality, cost and production norms set by the organisation.

Scope

The scope covers the following :

- Setting up machine
- Perform machining on the component

Elements and Performance Criteria

Setting up machine

To be competent, the user/individual on the job must be able to:

- PC1.** adjust the machine controls to ensure conformance with the specified tolerances
- PC2.** mount, install and align tools, attachments and fixtures on machine by using hand tools and precision measuring instruments
- PC3.** lift the work piece/metal stock manually or by hoist, position the same securely on the machine bed by using work holding devices and verify their positions with measuring instruments if required

Perform machining on the component

To be competent, the user/individual on the job must be able to:

- PC4.** cut the metal blocks into required size by using power operated/manual/automatic cutting tools as per the requirement
- PC5.** start the machine for machining operations
- PC6.** ensure that the right programme is selected in the CNC machine as defined in the SOP
- PC7.** perform various machining operations on the metal block to get the shape and dimension as per the drawing/work order
- PC8.** cut, shape and trim the metal block to specified lengths and shapes by using conventional/CNC machines
- PC9.** monitor the process parameters by reading the various gauges and correct them if not within standards
- PC10.** monitor the machine operations for any malfunctions/defects in the component and inform the supervisor/maintenance team for correction
- PC11.** measure the machined pieces and compare with the dimensions as prescribed in the work order and engineering drawing
- PC12.** follow the do's and don'ts of the manufacturing process as defined in SOPs/Work Instructions or given by supervisors
- PC13.** escalate the problems faced during machining to the supervisor

Knowledge and Understanding (KU)



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The individual on the job needs to know and understand:

- KU1.** relevant standards and procedures followed in the company
- KU2.** impact of not following defined procedures/ work instructions
- KU3.** various types of machining processes such as drilling, boring, turning etc.
- KU4.** SOP recommended by the manufacturer for using tools, jigs, fixtures, measuring instruments etc., required during the fabrication work
- KU5.** fundamentals of the CNC/conventional machine
- KU6.** SOP recommended by the organisation for operating CNC and conventional machine
- KU7.** safety requirements during the fabrication work

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read signals, work instructions, equipment manuals and process documents
- GS2.** communicate the process requirements to the supervisor and co-workers
- GS3.** attentively listen and comprehend the information given by the supervisor/team members
- GS4.** write work related information in English/regional language
- GS5.** recognise a workplace problem and take suitable action
- GS6.** analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently
- GS7.** plan and organise work according to the work requirements
- GS8.** complete the assigned tasks with minimum supervision
- GS9.** report to the supervisor or deal with a colleague individually, depending on the type of concern



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Setting up machine</i>	7	11	-	6
PC1. adjust the machine controls to ensure conformance with the specified tolerances	2	3	-	2
PC2. mount, install and align tools, attachments and fixtures on machine by using hand tools and precision measuring instruments	3	5	-	3
PC3. lift the work piece/metal stock manually or by hoist, position the same securely on the machine bed by using work holding devices and verify their positions with measuring instruments if required	2	3	-	1
<i>Perform machining on the component</i>	23	39	-	14
PC4. cut the metal blocks into required size by using power operated/manual/automatic cutting tools as per the requirement	2	3	-	2
PC5. start the machine for machining operations	1	2	-	-
PC6. ensure that the right programme is selected in the CNC machine as defined in the SOP	2	3	-	2
PC7. perform various machining operations on the metal block to get the shape and dimension as per the drawing/work order	6	10	-	4
PC8. cut, shape and trim the metal block to specified lengths and shapes by using conventional/CNC machines	4	7	-	3
PC9. monitor the process parameters by reading the various gauges and correct them if not within standards	2	3	-	1
PC10. monitor the machine operations for any malfunctions/defects in the component and inform the supervisor/maintenance team for correction	1	2	-	-
PC11. measure the machined pieces and compare with the dimensions as prescribed in the work order and engineering drawing	3	5	-	1



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. follow the do's and don'ts of the manufacturing process as defined in SOPs/Work Instructions or given by supervisors	1	2	-	1
PC13. escalate the problems faced during machining to the supervisor	1	2	-	-
NOS Total	30	50	-	20



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0902
NOS Name	Perform fabrication activities
Sector	Iron and Steel
Sub-Sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory, Iron Making
Occupation	Mechanical Maintenance
NSQF Level	3
Credits	TBD
Version	2.0
Last Reviewed Date	01/04/2025
Next Review Date	01/04/2028
NSQC Clearance Date	01/04/2025



Qualification Pack

ISC/N0903: Perform post-fabrication activities

Description

This unit is about performing post - fabrication activities such as inspection, quality check, cleaning etc.

Scope

The scope covers the following :

- Perform de- burring activity on the machined components
- Quality check of the machined component
- Carry out housekeeping and documentation

Elements and Performance Criteria

Perform de- burring activity on the machined components

To be competent, the user/individual on the job must be able to:

- PC1.** remove the workpieces from the machine as per company procedure
- PC2.** conduct de-burring operations with the help of correct tool to remove extra burrs, sharp edges, rust and chips from the metal surface
- PC3.** conduct shot blasting/vibro processes for completing de-burring operations
- PC4.** clean machine parts as per the defined process and quality control standards

Quality check of the machined component

To be competent, the user/individual on the job must be able to:

- PC5.** check the component as per the control plan, WI for product quality
- PC6.** ensure use of calibrated equipment to check the workpiece for conformance to the required specifications and standards
- PC7.** note down the observations of the basic inspection process and identify pieces which are as per the specified standards
- PC8.** separate the completed pieces into Ok pieces and defective pieces which can be repaired/reworked and pieces which are beyond repair and maintain records of each category

Carry out housekeeping and documentation

To be competent, the user/individual on the job must be able to:

- PC9.** clean and store all the tools, machine and equipment after completion of work
- PC10.** dispose scrap or waste material into the disposal area in accordance with the company's policies and environmental regulations
- PC11.** maintain and update all the records and reports related to fabrication work done as per the organisational guidelines

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:



Qualification Pack

- KU1.** Standard Operating Procedures (SOP) recommended by the manufacturer for using tools and measuring instruments used after the machining processes
- KU2.** the post machining processes like deburring, cleaning, quality check etc.
- KU3.** the impact of presence of burrs, edges, chips on the final product quality
- KU4.** the various inspection methods for inspecting the quality of machined product
- KU5.** SOP recommended by the organisation for checking defects in the product/work piece
- KU6.** various type of defects in final products
- KU7.** impact of non-conformities on final product and their causes to quality assurance standards
- KU8.** escalation matrix for reporting identified problem
- KU9.** records to be maintained and implications of non-maintenance of the same

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read work instructions, equipment manuals and process documents
- GS2.** communicate the process requirements to the supervisor and co-workers
- GS3.** attentively listen and comprehend the information given by the supervisor/team members
- GS4.** write work related information in English/regional language
- GS5.** recognise a workplace problem and take suitable action
- GS6.** analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently
- GS7.** plan and organise work according to the work requirements
- GS8.** complete the assigned tasks with minimum supervision
- GS9.** report to the supervisor or deal with a colleague individually, depending on the type of concern



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform de-burring activity on the machined components</i>	12	20	-	8
PC1. remove the workpieces from the machine as per company procedure	2	3	-	1
PC2. conduct de-burring operations with the help of correct tool to remove extra burrs, sharp edges, rust and chips from the metal surface	4	7	-	3
PC3. conduct shot blasting/vibro processes for completing de-burring operations	4	7	-	3
PC4. clean machine parts as per the defined process and quality control standards	2	3	-	1
<i>Quality check of the machined component</i>	11	19	-	8
PC5. check the component as per the control plan, WI for product quality	4	8	-	3
PC6. ensure use of calibrated equipment to check the workpiece for conformance to the required specifications and standards	2	3	-	2
PC7. note down the observations of the basic inspection process and identify pieces which are as per the specified standards	2	3	-	1
PC8. separate the completed pieces into Ok pieces and defective pieces which can be repaired/reworked and pieces which are beyond repair and maintain records of each category	3	5	-	2
<i>Carry out housekeeping and documentation</i>	7	11	-	4
PC9. clean and store all the tools, machine and equipment after completion of work	3	5	-	2
PC10. dispose scrap or waste material into the disposal area in accordance with the company's policies and environmental regulations	2	3	-	1
PC11. maintain and update all the records and reports related to fabrication work done as per the organisational guidelines	2	3	-	1



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	50	-	20



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0903
NOS Name	Perform post-fabrication activities
Sector	Iron and Steel
Sub-Sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory, Iron Making
Occupation	Mechanical Maintenance
NSQF Level	3
Credits	TBD
Version	2.0
Last Reviewed Date	01/04/2025
Next Review Date	01/04/2028
NSQC Clearance Date	01/04/2025



Qualification Pack

ISC/N0904: Perform maintenance activities

Description

This unit is about performing maintenance of equipment used for fabrication

Scope

The scope covers the following :

- Perform preventive maintenance of machines

Elements and Performance Criteria

Perform preventive maintenance of machines

To be competent, the user/individual on the job must be able to:

- PC1.** identify and arrange necessary inspection and repairing tools required during maintenance work
- PC2.** observe the machines or equipment to diagnose machine malfunction and determine need for adjustment or repair
- PC3.** maintain the machine as per proper operational condition/daily maintenance check list
- PC4.** clean and oil the machine and its components as per checklist
- PC5.** clean the hydraulic tank/gauge/tools/fixtures as per the cleaning schedule provided in Work Instruction/SOP manual
- PC6.** check coolant and lubricant level in the machine as per standards
- PC7.** apply appropriate lubricant as per manufacturer specification
- PC8.** remove chips from different machine areas and dispose scrap or waste material into the disposal area in accordance with the company policies and environmental regulations
- PC9.** carry out minor repairs and adjustments of the machine and report any malfunctions/repairs in the machine beyond own scope to the concerned person
- PC10.** replace worn out tools timely and safely with new tools
- PC11.** maintain and update all the records and reports related to maintenance work done as per the organisational guidelines

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** checks to be done during preventive maintenance and importance of the same
- KU2.** types of documentation in organization and importance of the same
- KU3.** records to be maintained and implications of non-maintenance of the same
- KU4.** procedure of assemble/ dis-assemble an equipment with a given design
- KU5.** use of equipment such as thermometer, tachometer and vibration monitoring tools to perform preventive maintenance



Qualification Pack

- KU6.** normal running characteristics of machines
- KU7.** basic troubleshooting and repairing of machines
- KU8.** potential problems with machines, their causes and remedies

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read work instructions, equipment manuals and process documents
- GS2.** communicate the process requirements to the supervisor and co-workers
- GS3.** attentively listen and comprehend the information given by the supervisor/team members
- GS4.** write work related information in English/regional language
- GS5.** recognise a workplace problem and take suitable action
- GS6.** analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently
- GS7.** plan and organise work according to the work requirements
- GS8.** complete the assigned tasks with minimum supervision
- GS9.** report to the supervisor or deal with a colleague individually, depending on the type of concern



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform preventive maintenance of machines</i>	30	50	-	20
PC1. identify and arrange necessary inspection and repairing tools required during maintenance work	4	4	-	2
PC2. observe the machines or equipment to diagnose machine malfunction and determine need for adjustment or repair	3	5	-	2
PC3. maintain the machine as per proper operational condition/daily maintenance check list	2	3	-	1
PC4. clean and oil the machine and its components as per checklist	2	5	-	2
PC5. clean the hydraulic tank/gauge/tools/fixtures as per the cleaning schedule provided in Work Instruction/SOP manual	3	5	-	2
PC6. check coolant and lubricant level in the machine as per standards	3	5	-	2
PC7. apply appropriate lubricant as per manufacturer specification	3	5	-	2
PC8. remove chips from different machine areas and dispose scrap or waste material into the disposal area in accordance with the company policies and environmental regulations	3	5	-	2
PC9. carry out minor repairs and adjustments of the machine and report any malfunctions/repairs in the machine beyond own scope to the concerned person	3	7	-	3
PC10. replace worn out tools timely and safely with new tools	2	3	-	1
PC11. maintain and update all the records and reports related to maintenance work done as per the organisational guidelines	2	3	-	1
NOS Total	30	50	-	20



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0904
NOS Name	Perform maintenance activities
Sector	Iron and Steel
Sub-Sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory
Occupation	Mechanical Maintenance
NSQF Level	3
Credits	TBD
Version	2.0
Last Reviewed Date	01/04/2025
Next Review Date	01/04/2028
NSQC Clearance Date	01/04/2025



Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team



Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services



Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.
6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.



Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ISC/N0008.Use basic health and safety practices at the workplace	30	50	0	20	100	5
ISC/N0009.Work effectively with others	30	50	0	20	100	5
ISC/N0901.Prepare for fabrication activities	30	50	-	20	100	15
ISC/N0902.Perform fabrication activities	30	50	-	20	100	25
ISC/N0903.Perform post-fabrication activities	30	50	-	20	100	25
ISC/N0904.Perform maintenance activities	30	50	-	20	100	15
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	0	0	50	10
Total	200	330	-	120	650	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
5S	Technique of maintaining orderliness “Japanese terminology.
CP	Control Plan.
WI	Work Instructions.



Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.